# The Exchange Phase as a Logical Scalar: $R = e^{2\pi i s}$ from the Re-Entrant Calculus

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2026-08-16

## Abstract

The boson/fermion dichotomy is conventionally presented as a primitive classification of nature. Recent work established that the underlying invariant is the relation between the exchange phase of identical particles and their topological spin, $R = e^{2\pi i s}$, with the dichotomy as its shadow in three spatial dimensions where the involutive braiding quantizes the spin parameter. This paper examines whether a calculus whose only primitive is the act of drawing a distinction can generate this invariant rather than importing it as an axiom. The re-entrant mark of the calculus of indications, disciplined by linearity, already generates the constants e and π as logical scalars: e as the fixed point of the differential equation $D f = f$, and π as the trace of the identity on the circle type. The central observation of this paper is that the exchange phase is the (2s)-fold half-turn of the re-entrant mark: $R = (e^{i\pi})^{2s} = e^{2\pi i s} = (-1)^{2s}$, so that the boson/fermion dichotomy is the parity of 2s. The arithmetic identity is established; the identification of the exchange monodromy with a power of the mark's half-turn is a model of the re-entrant phase; the claim that the calculus derives the invariant as a logical scalar within a single formal system is stated as a conjecture with explicit falsifiability conditions. The framework is distinguished from the nearest prior work deriving a $Z_2$ exchange phase from self-referential scattering, which lacks both the power structure for arbitrary spin and the unification of e, π, and R as a single scalar family.

**Keywords:** exchange phase; spin-statistics; topological spin; laws of form; re-entrant mark; logical scalar; anyons

## 1. Introduction

The spin-statistics theorem is among the most robust laws of physics: integer-spin particles obey Bose–Einstein statistics and half-integer-spin particles obey Fermi–Dirac statistics [Pauli 1940]. The QNFO research program has pursued the structural reading of this law. In [Quni-Gudzinas 2026b] it was established that the primitive content is not the dichotomy itself but the relation

$$R = e^{2\pi is} \quad \text{[ESTABLISHED]}$$

between the exchange phase of identical particles and their topological spin, where the boson/fermion binary is a dimension-dependent shadow of this relation in three spatial dimensions [ESTABLISHED]. The same work identified a derivation target: a distinction-based foundation of physics must derive exchange statistics from the primitive act of drawing a boundary, rather than importing the relation as an axiom. The required construction was formalized — two modal exponentials, the braiding of two marks in a compact closed category, and the ribbon condition — yielding η = ±1 in the symmetric case, with η = −1 identified with the treatise's half-turn phase e^{iπ} = −1.

This paper closes the remaining gap with one structural observation: **the exchange phase is the (2s)-fold half-turn of the re-entrant mark**. The constants e and π have already been shown to arise as logical scalars of the re-entrant calculus — values the type theory computes rather than axioms it imports [Quni-Gudzinas 2026a]. This paper extends the scalar family to the exchange phase R = (e$^{\{i\pi\})}${2s}, and states precisely what is established, what is a model, and what is conjectural about that identification.

# 2. So What? Why Should a Reader Care About This Research?

**The stakes.** The spin-statistics theorem is among the most robust laws of physics: integer-spin particles obey Bose–Einstein statistics and half-integer-spin particles obey Fermi–Dirac statistics [Pauli 1940]. Yet the standard derivation imports Lorentz invariance, microcausality, and positive energy — and it takes the *form* of the exchange phase for granted. The constants e and π enter physics as brute inputs. This paper asks whether they are outputs: if the exchange phase is the (2s)-fold half-turn of the re-entrant mark, then e, π, and R = e^{2πis} form a single scalar family generated by the act of drawing a distinction. The two most famous constants in physics would not be axioms of nature but consequences of a more primitive logical act.

**How deep does the theorem go?** The premises-depth audit of Section 6 separates what is established from what is assumed: (a) the arithmetic identity R = e^{2πis} = (−1)^{2s} is established; (b) the identification of the exchange monodromy with a power of the mark's half-turn is a model of the re-entrant phase, not a theorem; (c) the full claim — that the calculus derives the invariant as a logical scalar within a single formal system — is a conjecture with explicit falsification conditions (Section 7). The reader is told exactly where the premises end.

**Why should a physicist care?** A first-principles account of the exchange phase's *form* is a classification instrument: if the phase is a logical scalar, the space of possible statistics is constrained by the logical origin of the phase, not only by the topology of configuration space. The companion boundary map (Configuration-Space Topology and the Distinction Calculus, QNFO) already converts this into a platform map for topological quantum computation — which framework conditions generate which anyon sectors (orbifolds, traid groups, graph configuration spaces) — and the braiding-phase-gate roadmap (bp-gates.md, deposited with this record) is the concrete instrument: gate sets generated from braid words classified by the calculus, hardware-agnostic, with a falsifiable prediction for each application.

**Why should a mathematician or foundations reader care?** The claim that e and $\pi$ arise as logical scalars — e as the fixed point of the differential equation $Df = f$, $\pi$ as the trace of the identity on the circle type — is falsifiable, and the falsification conditions F1–F3 (Section 7) give concrete tests. This is the premises-depth question asked of the most robust law in physics: not "is the law true?" but "what is the law an output of?"

**Practical utility — even if the conjecture fails.** The value does not depend on the conjecture's success. (1) The boundary map of what the calculus cannot derive — the spin-statistics *connection* still requires Lorentz input — is itself the deliverable. (2) The arithmetic identity $R = (e^{i\pi})^{2s} = (-1)^{2s}$ is a reusable instrument for classifying exchange phases. (3) The logical-loop simulation program (iterating re-entrant marks without solving the Schrödinger equation) gives an independent kinematic check of anyon models. (4) The braiding-phase-gate roadmap (bp-gates.md) is a concrete application path toward topological quantum computation.

**What this paper does not claim.** No derivation of the spin-statistics connection (which spin belongs to which statistics) — that boundary is conceded. No claim that the calculus is the unique origin of e and $\pi$. No completed formal system: the derivation is stated with falsification conditions, and the status ladder (Section 6) grades every component honestly. What the reader gets is a falsifiable research program and a map of where the premises end — the answer to "how deep does this go?"

# 3. Background: the invariant and the machinery

## 3.1 The invariant $R = e^{2\pi is}$

The exchange of two identical particles is a loop in their configuration space. The phase acquired is a representation of the fundamental group of that space: $\pi_1 = \mathbb{Z}_2$ in dimension $d \geq 3$, and $\pi_1 = B_2 \cong \mathbb{Z}$ in dimension $d = 2$ [Leinaas and Myrheim 1977]. The spin-statistics relation states which phase is realized: $\eta = (-1)^{2s}$, where s is the spin [Pauli 1940; Duck and Sudarshan 1998]. In dimension $d \geq 3$ the involutive braiding forces $2s \in \mathbb{Z}$, so the exchange phase is $\pm 1$ and particles are bosons or fermions; in dimension $d = 2$ the quantization collapses to continuity, and arbitrary exchange phases $e^{2\pi is}$ describe anyons [Wilczek 1982; Kitaev 2006]. [ESTABLISHED]

## 3.2 The re-entrant calculus

The calculus of indications [Spencer-Brown 1969] takes the drawing of a distinction as primitive. The re-entrant form $f = \bar{f}$ oscillates, generating a discrete clock of period 2: $f(n) = (-1)^n f(0)$ [ESTABLISHED — elementary consequence]. Under linear discipline [Quni-Gudzinas 2026a]:

- **e** arises as the logical scalar of the differential fixed point: $D f = f$, $f(0) = 1$, with unique solution $f(x) = e^x$ and $e = \Sigma\, 1/n!$ [established analysis];
- ***π*** arises as the trace of the identity on the circle type: $\pi = \mathrm{Tr}(\mathrm{id}_{S^1}) = C/d$ in the analytic realization, where the circle carries its geometric structure and the trace is computed by the Gaussian-integral construction [established geometry; logical derivation: [my conjecture]];

- **the half-turn** of the circle carries the marked state to the unmarked state: e^{iπ} = −1, the geometric root of the Euler identity [ESTABLISHED — Euler's formula].

The exchange of two marks in a compact closed category was constructed in [Quni-Gudzinas 2026b, notebook T2]: the exchange map σ_{M,M} = η·id for the self-dual mark, the ribbon identity η = θ_M, and the symmetric-category constraint η = ±1, with η = −1 identified with Crossing (e^{iπ} = −1) and η = +1 with Calling.

### 3.3 The gap

The parent works establish η = ±1 (symmetric case) and identify the single sign η = −1 with the treatise's half-turn, but never write the power structure R = (e^{iπ})^{2s} that covers arbitrary spin s and the 2+1-dimensional anyon generalization. The leap from "the mark has a half-turn phase" to "two marks anticommute" is asserted, not derived [Quni-Gudzinas 2026b]. This paper supplies the missing composite reading.

# 4. Core claim

> **The exchange phase R = e^{2πis} is the (2s)-fold half-turn of the re-entrant mark:**
>
> $$R = (e^{i\pi})^{2s} = e^{2\pi i s} = (-1)^{2s},$$
>
> **with the boson/fermion dichotomy the parity of 2s.**

**Formally:** - s ∈ ℤ: 2s even → R = +1 → Bose–Einstein statistics (symmetric exchange, Calling); - s ∈ ℤ + ½: 2s odd → R = −1 → Fermi–Dirac statistics (antisymmetric exchange, Crossing); - 3+1-dimensional involutive braiding quantizes s to {0, 1/2, 1, 3/2, …} → the dichotomy [ESTABLISHED, Quni-Gudzinas 2026b]; - 2+1 dimensions allow any real s → anyon phases e^{2πis} [ESTABLISHED, Leinaas and Myrheim 1977].

**Scope note on the equality chain.** The final equality (−1)^{2s} = ±1 holds only in the quantized case 2s ∈ ℤ. For arbitrary real s the general form is R = e^{2πis} = cos 2πs + i sin 2πs (e.g., s = 1/4 gives R = i). The specific reading advanced here is the middle form: R as a power of the treatise's half-turn, (e^{iπ})^{2s} — the monodromy of the exchange loop as the (2s)-fold iteration of the half-turn e^{iπ} = −1.

# 5. The derivation

**Step 1 — the exchange map exists in the compact closed structure.** [established — construction] The exchange σ_{M,M}: M⊗M → M⊗M is the braiding of the self-dual mark. In a symmetric braided category $\sigma^2$ = id, so σ has eigenvalues ±1 with idempotent projectors P_sym = ½(1+σ) and P_antisym = ½(1−σ).

**Step 2 — the half-turn as the basic monodromy.** [established — treatise §12.1] The half-turn of the circle carries the marked state to the unmarked state: e^{iπ} = −1 = Crossing. This is the single-mark monodromy under rotation by π.

**Step 3 — exchange phase as a power of the half-turn.** [MAP — model of the re-entrant phase] The exchange of two particles is a loop in their configuration space. The loop winds the relative coordinate; the phase acquired is the monodromy of that loop. The reading advanced here is that the exchange monodromy is the (2s)-fold iteration of the mark's half-turn: $R = (e^{i\pi})^{2s} = e^{2\pi i s} = (-1)^{2s}$. The arithmetic identity is [established]; the identification of the exchange monodromy with a power of the mark's half-turn is [MAP — model].

**Step 4 — parity of 2s → the dichotomy.** [established arithmetic + Quni-Gudzinas 2026b §1] The dichotomy is exactly the parity of 2s.

**Step 5 — dimension quantization.** [established — Quni-Gudzinas 2026b §3] In $d \geq 3$ the involutive braiding forces $2s \in \mathbb{Z} \rightarrow R = \pm 1$; in $d = 2$ the braid group allows continuous $s \rightarrow$ anyon phases $e^{2\pi i s}$.

**Step 6 — unification with e and π.** [my conjecture — the scalar family] e (the fixed point of $D f = f$), π (the trace of the identity on $S^1$), and R (the monodromy power of the half-turn) form one family of logical scalars of the re-entrant mark under linear discipline: fixed point, trace, monodromy power.

**Formalization (P4, 2026-08-16).** A formal derivation of the composite in the Part VIII system — the traced differential cohesive linear type theory of the treatise (§34–§36) — is provided in the companion artifact `artifacts/p4-formal-derivation.md`: the composite $(e^{i\pi})^{2s}$ is expressible as the (2s)-fold composition of the half-turn endomorphism of $S^1$, an element of $\mathrm{End}(S^1)$ (the phase rotations, §34.1), evaluated via the trace/scalar structure (§34.2, §36.1). Every status label in the derivation is audited to its floor: the trace/compact-closed machinery and $S^1$ self-duality are [established] (Joyal–Street–Verity; HoTT); the half-turn $e^{i\pi} = -1$ is [established] (§12.1, Euler's formula); the exchange-monodromy identification is [MAP — model]; the axiom-free computation of the constants e and π by the bare type-theoretic syntax is [my conjecture]. **F1 is thereby partially discharged:** the construction is complete within established components plus the declared MAP; the residual open step is the Appendix D proof-assistant computation, registered in the research-continuity-registry (FQ1) — not claimed.

# 6. Status ladder

| Component | Status |
|---|---|
| Exchange map $\sigma_{M,M}$; projectors $P_{sym}/P_{antisym}$ | [established — Quni-Gudzinas 2026b, notebooks T1/T2] |
| Half-turn $e^{i\pi} = -1$ | [established — treatise §12.1] |
| $\eta = -1 \leftrightarrow$ Crossing; $\eta = +1 \leftrightarrow$ Calling | [established — Quni-Gudzinas 2026b, notebook T2] |
| $(e^{i\pi})^{2s} = (-1)^{2s} = \pm 1$ for $2s \in \mathbb{Z}$ | [established — elementary arithmetic] |
| Exchange monodromy = (2s)-fold half-turn | [MAP — model of the re-entrant phase] |
| e/π/R scalar-family unification | [my conjecture] |
| Formal derivation in the traced differential cohesive linear type theory | [my conjecture — F1 target; PARTIALLY DISCHARGED 2026-08- |

| Component | Status |
|---|---|
| | 16: construction delivered in p4-formal-derivation.md; axiom-free computation of the constants = Appendix D path, open] |
| Physical realization (which sign, in 3+1D) | [established physics; external Lorentz/microcausality input] |

# 7. Falsifiability conditions

- **F1 (formal).** The claim that the re-entrant calculus *generates* $R = (e^{i\pi})^{2s}$ as a logical scalar is falsified if no derivation exists within the traced differential cohesive linear type theory of the treatise (Part VIII) without importing the relation as an axiom. This is a concrete, checkable claim about a formal system. [my conjecture]
- **F2 (empirical, inherited).** If a stable, local, relativistic excitation in 3+1 dimensions is observed with exchange phase $\eta \neq e^{2\pi i s}$ (e.g., a spin-1/2 particle obeying Bose–Einstein statistics), the invariant claim is disconfirmed. No such particle is known in the Standard Model. Evasion strategies in the literature (e.g., mass-dimension-three-half spinors [Ahluwalia and Lee 2022]) target the standard theorem statement rather than the invariant relation itself. [established — restated from Quni-Gudzinas 2026b F1]
- **F3 (scope).** The arithmetic identity $R = (e^{i\pi})^{2s} = (-1)^{2s}$ is [established] elementary arithmetic; the identification of the exchange phase with the (2s)-fold half-turn in the geometric model is [MAP — model]; the claim that this identification is a logical derivation within a single formal system is [my conjecture].

**Disconfirmation summary.** The invariant claim is disconfirmed if: (a) a formal derivation in the Part VIII system is shown impossible without importing the relation as an axiom (F1); or (b) an empirical excitation with exchange phase $\eta \neq e^{2\pi i s}$ is observed in 3+1 dimensions (F2). Neither condition is currently met.

# 8. Relation to prior work

**Quni-Gudzinas 2026b (parent).** Establishes the invariant $R = e^{2\pi i s}$ and the $\eta = \pm 1$ symmetric construction; explicitly documents that the leap from the half-turn phase to anticommuting marks is not derived. This paper supplies the $(e^{i\pi})^{2s}$ composite reading absent there.

**Kauffman 2022.** Reviews Majorana fermions through the laws of form, connecting the mark calculus to fermionic structure. It addresses the representation of fermions, not the exchange-phase invariant, and contains no (2s)-fold half-turn power structure. The present claim is distinct and complementary.

**Kauffman 2013 (World Scientific).** “Laws of Form, Majorana Fermions, and Discrete Physics” (*The Physics of Reality*, pp. 1–18) is the direct predecessor of the 2022 review, connecting the mark calculus to Majorana structure a decade earlier. It likewise does not construct the (2s)-fold half-turn power structure nor the exchange-phase invariant; the present claim remains distinct.

**Kauffman 2013 (arXiv:1301.6214).** "Knot Logic and Topological Quantum Computing with Majorana Fermions" develops the mark-as-fermion-algebra reading: negation seen as the mark "naturally generates the fermion algebra, the quaternions and the braid group representations related to Majorana fermions." This is the closest prior art on the mark–fermion identification. The distinction maintained here: Kauffman derives fermion *algebra* (Clifford/quaternion structure) from the mark; the present work derives the exchange-phase *invariant* R = $(e^{\{i\pi\}})\{2s\}$ as a monodromy-power logical scalar. Algebra vs invariant is the novelty boundary (P3 classification, support-4).

**Ma and Zhang 2025.** Derive a $Z_2$ exchange phase from self-referential scattering via Riccati square roots and the spinor double cover in a quantum field theory framework. Their primitive is self-referential scattering in QFT; the primitive here is the re-entrant mark of the calculus of indications. Their result is confined to the $Z_2$ (boson/fermion) case; the $(e^{\{i\pi\}})\{2s\}$ power structure for arbitrary s (anyons) and the e/π/R scalar-family unification are absent. The present claim is distinct: the derivation target is the re-entrant mark under linear discipline, not bare self-reference.

**Berry and Robbins 2017.** The geometric-phase construction of spin-statistics: exchange of two particles acquires the Berry phase, connecting statistics to geometry. The monodromy-power reading advanced here is conceptually adjacent (both trace the exchange phase to a geometric monodromy) but is formulated natively in the calculus of indications rather than in Hilbert-space geometry.

**Ahluwalia and Lee 2022.** Propose mass-dimension-three-half spinors as an evasion of the standard spin-statistics theorem. This is relevant to F2: the evasion targets the standard theorem statement, not the invariant R itself; the empirical falsifier above covers the general evasion class.

**Sato and Fujimoto 2016.** The canonical condensed-matter review of Majorana fermions and topology in superconductors (*J. Phys. Soc. Jpn.* 85, 072001). Provides the physics context for the Majorana connection; contains no Laws-of-Form content (Background).

**Vissani 2026.** History-of-physics reconstruction of Majorana's 1933–1937 route to anti-commuting quantization (arXiv:2603.28538). Historical anchor; no Laws-of-Form content (Background).

**Kauffman 1980/1995/2023.** The Laws-of-Form foundations corpus — "Form dynamics" (*J. Social Biol. Syst.* 3(2), 1980), "Arithmetic in the Form" (*Cybernetics and Systems* 26(1), 1995), "Autopoiesis and Eigenform" (*Computation* 11(12), 2023) — anchors the reading of the calculus used throughout this paper.

# 9. Conclusions

The exchange phase R = e^{2πis} is the (2s)-fold half-turn of the re-entrant mark: R = $(e^{\{i\pi\}})\{2s\}$ = (−1)^{2s}. The boson/fermion dichotomy is the parity of 2s. The claim is scoped with an explicit status ladder: the arithmetic is established, the identification is a model, and the full logical derivation within the traced differential cohesive linear type theory of the treatise is a conjecture with concrete falsifiability conditions. If the derivation succeeds, e, π, and R form a single family of logical scalars of the re-entrant mark — fixed point, trace, and monodromy power — and the spin-statistics connection is the arithmetic of the half-turn. If it fails, the failure mode is precisely specified: the calculus cannot generate the exchange phase without importing it as an axiom.

# 10. Declarations


- **Funding:** This research received no external funding.

- **Conflicts of interest:** The author declares no conflicts of interest.
- **Data availability:** All source files, gate artifacts, and external-search evidence are deposited with this record and mirrored in the project repository (see provenance link).
- **Code availability:** The derivation is analytical; the arithmetic verification and citation-audit scripts are deposited as artifacts.
- **Ethics approval:** Not applicable.
- **Consent for publication:** Not applicable.
- **Author contributions:** R.B.Q.-G. conceived, derived, and wrote the paper.
- **Preprint policy:** This is a self-archived working paper; it has not been submitted for peer review.
- **Reproducibility:** Every numerical claim is independently recomputable from the stated elementary formulas; the citation audit re-verifies every bibliographic entry against live registries.

# References


- Ahluwalia, D. V., and C.-Y. Lee (2022). Spin-half bosons with mass dimension three-half: Evading the spin-statistics theorem. *Europhysics Letters*, 10.1209/0295-5075/ac97bd (+ erratum 10.1209/0295-5075/acabe2).
- Berry, M. V., and J. M. Robbins (2017). Indistinguishability for quantum particles: spin, statistics and the geometric phase. *A Half-Century of Physical Asymptotics and Other Diversions*, 10.1142/9789813221215_0008.
- Duck, I., and E. C. G. Sudarshan (1998). Toward an understanding of the spin-statistics theorem. *American Journal of Physics* 66(4), 284–303, 10.1119/1.18860.
- Kauffman, L. H. (2022). A Review of Majorana fermions and the laws of form. *Journal of Physics: Conference Series* 2197, 012001, 10.1088/1742-6596/2197/1/012001.
- Kitaev, A. (2006). Anyons in an exactly solved model and beyond. *Annals of Physics* 321(1), 2–111, 10.1016/j.aop.2005.10.005.
- Leinaas, J. M., and J. Myrheim (1977). On the theory of identical particles. *Il Nuovo Cimento B* 37, 1–23, 10.1007/BF02727953.
- Ma, H., and W. Zhang (2025). Self-Referential Scattering and the Birth of Fermions: Riccati Square Roots, Spinor Double Cover, and a $Z_2$ Exchange Phase. Zenodo, 10.5281/zenodo.17706898.
- Pauli, W. (1940). The Connection Between Spin and Statistics. *Physical Review* 58, 716, 10.1103/PhysRev.58.716.
- Quni-Gudzinas, R. B. (2026a). The Calculus of Re-Entrant Distinctions: A Unified Treatise on the Loop, the Tree, and the Constants of Self-Reference.

Zenodo, 10.5281/zenodo.21908818.

- Quni-Gudzinas, R. B. (2026b). The Boson/Fermion Distinction: Spin-Statistics as Structural Invariant. Zenodo, 10.5281/zenodo.21938971.
- Spencer-Brown, G. (1969). *Laws of Form*. George Allen and Unwin.
- Wilczek, F. (1982). Magnetic Flux, Angular Momentum, and Statistics. *Physical Review Letters* 48, 1144, 10.1103/PhysRevLett.48.1144.
- Joyal, A., R. Street, and D. Verity (1996). Traced monoidal categories. *Mathematical Proceedings of the Cambridge Philosophical Society* 119, 447–468, 10.1017/S0305004100074338.
- Kauffman, L. H. (1980). Form dynamics. *Journal of Social and Biological Systems* 3(2), 171–206, 10.1016/0140-1750(80)90008-1.
- Kauffman, L. H. (1995). Arithmetic in the Form. *Cybernetics and Systems* 26(1), 1–57, 10.1080/01969729508927486.
- Kauffman, L. H. (2013). Laws of Form, Majorana Fermions, and Discrete Physics. *The Physics of Reality*, World Scientific, 1–18, 10.1142/9789814504782_0001.
- Kauffman, L. H. (2013). Knot Logic and Topological Quantum Computing with Majorana Fermions. arXiv:1301.6214.
- Kauffman, L. H. (2023). Autopoiesis and Eigenform. *Computation* 11(12), 247, 10.3390/computation11120247.
- Sato, M., and S. Fujimoto (2016). Majorana Fermions and Topology in Superconductors. *Journal of the Physical Society of Japan* 85, 072001, 10.7566/JPSJ.85.072001.
- Vissani, F. (2026). From Hole Theory to Quantum Field Theory: Relativistic Fermions and the Role of Ettore Majorana (1933–1937). arXiv:2603.28538.